Gamification Domain Ontology - Core Concepts (GaDO-Core)

Important Note: The guidelines presented on this document are based on the METHONTOLOGY methodology.

1. Specification

Domain: Gamification
Date: Sep 15, 2015
Level of formality: Semi-formal.

Conceptualized and Implemented by: Diego Dermeval

Purpose: Ontology about gamification concepts. This ontology will be used on a computer-based educational context, however, it does not limit itself to a single context and it can be generalized accordingly.

Scope: Game design, gamification concepts, game mechanics, psychological and need theories, player models.

Sources of Knowledge:

- Werbach, Kevin, and Dan Hunter. *For the win: How game thinking can revolutionize your business*. Wharton Digital Press, 2012.
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2. Knowledge Acquisition

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- Game design

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Towards an Ontology for Gamifying Collaborative Learning Scenarios

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2.2. References from: Gamification in theory and action: A survey

- Game interface design patterns [Deterding et al. (2011a, p. 12).]
    - Badge, Leaderboard, Level
- Game design patterns and mechanics
    - Time constraint, Limited resources, Turns
- Game design principles and heuristics
    - Enduring play, Clear goals, Variety of game styles
- Game models
    - MDA; challenge, fantasy, curiosity, Game design atoms; Core elements of the gaming experience (CEGE)
- Game design methods
    - Playtesting, playcentric design, value conscious game design
- Frameworks
    - Aparicio et. al (2012), Nicholson (2012), Sakamoto et al. (2012), Blohm and Leimeister (2013)
    - Elements by Autonomy [Aparicio et. al (2012)]
        - Profiles, avatars, macros, configurable interface, alternative activities, privacy control, notification control
    - Elements by Competence [Aparicio et. al (2012)]
        - Positive feedback, optimal challenge, progressive information, intuitive controls, points, levels, leaderboards
    - Elements by Relation [Aparicio et. al (2012)]
        - Groups, messages, blogs, connection to social networks, chat.
- Game mechanics [Blohm and Leimeister (2013).]

    - Documentation of behavior, Scoring systems, badges, trophies, rankings, levels, reputation points, group tasks, time pressure, tasks, quests, avatars, virtual worlds, virtual trade
- Game elements according to Seaborn
    - Points (exp points, score), badges (trophies), leaderboards (rankings scoreboard), progression (levelling, level up), status (title, ranks), levels (stage, area, world), rewards (incentives, prizes, gifts), roles (class, character).
- Other Elements
    - Progress bar, milestones, missions, mini-games, hidden rewards, team

2.3. References from: For The Win

- Gamification types
    - Internal gamification, External gamification, Behavior-Change gamification
- Breaking down gamification:
    - Game Elements,
    - Game-design techniques
- Magic Circle
- **Dynamics**
    - Constraints Constraints (limitations or forced trade-offs)
    - Emotions (curiosity, competitiveness, frustration, happiness, and so on)
    - Narrative (a consistent, ongoing storyline)
    - Progression (the player's growth and development)
    - Relationships (social interactions generating feelings of camaraderie, status, altruism, and so on)
- **Mechanics**
    - Challenges (puzzles or other tasks that require effort to solve)
    - Chance (elements of randomness)
    - Competition (one player or group wins, and the other loses . . .)
    - Cooperation (players must work together to achieve a shared goal)
    - Feedback (information about how the player is doing)
    - Resource Acquisition (obtaining useful or collectible items)
    - Rewards (benefits for some action or achievement)
    - Transactions (trading between players, directly or through intermediaries)
    - Turns (sequential participation by alternating players)
    - Win States (objectives that makes one player or group the winner—draw and loss states are related concepts)
- **Components**
    - Achievements (defined objectives)
    - Avatars (visual representations of a player's character)
    - Badges (visual representations of achievements)
    - Boss Fights (especially hard challenges at the culmination of a level)
    - Collections (sets of items or badges to accumulate)
    - Combat (a defined battle, typically short-lived)

    - Content Unlocking (aspects available only when players reach objectives)
    - Gifting (opportunities to share resources with others)
    - Leaderboards (visual displays of player progression and achievement)
    - Levels (defined steps in player progression)
    - Points (numerical representations of game progression)
    - Quests (predefined challenges with objectives and rewards)
    - Social Graphs (representation of players' social network within the game)
    - Teams (defined groups of players working together for a common goal)
    - Virtual Goods (game assets with perceived or real-money value)
- Easy fun, Hard fun

2.4. References from: [Does Gamification Work? — A Literature Review of Empirical Studies on Gamification (Hamari)]

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    - [31] K. Huotari, and J. Hamari, "Defining gamification: a service marketing perspective", In Proceedings of the 16th International Academic MindTrek Conference, October 3-5, 2012, Tampere, Finland, ACM, pp. 17-22.
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- Leaderboards
    - [4][10][13][15][16][21][23][27][37][41]
- Achievements/Badges
    - [2][8][10][17][20][22][25][27][34]
- Levels
    - [11][15][16][21][27][37]
- Story/Theme
    - [12][18][21][23][33][36]
- Clear goals
    - [11][27][33][32]
- Feedback
    - [4][11][21][27][32][33]
- Rewards

    - [12][18][33][36]
- Progress
    - [14][18][27][33]
- Challenge
    - [4][13][18][21][23][27][33]
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Personal Growth	[44] Deborah A. Gears. Corporate wiki conduct: A study of organizational in uences, emotion, and motivation. Journal of Leadership, Accountability and Ethics, 9(3): 75{85, 2012.
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● Intrinsic motivators ○ altruism ○ competition ○ cooperation ○ sense of belonging ○ love ○ aggression	[112] Cristina Ioana Muntean. Raising engagement in e-learning through gamication. In Proc. 6th International Conference on Virtual Learning ICVL, pages 323{329, 2011.
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● Skinner's reinforcement theory ○ Reinforcement ○ Punishment	[134] B. F. Skinner. About Behaviorism. Vintage, New York, 1 edition edition, February 1976. ISBN 9780394716183.
● Csikszentmihalyi's Flow Theory ○ Challenge	[25] Mihaly Csikszentmihalyi. Learning, \ ow," and happiness. In Applications of

○ Ability ○ Anxiety ○ Boredom ○ Flow	Flow in Human Development and Education, pages 153{172. Springer Netherlands, January 2014. ISBN 978-94-017-9093-2, 978-94-017-9094-9. [136] C. R. (Richard) Snyder, Shane J. Lopez, and Jennifer T. Pedrotti. Positive Psy- chology: The Scientic and Practical Explorations of Human Strengths. SAGE Publications, Inc, Thousand Oaks, second edition edition edition, September 2010. ISBN 9781412990622. [160] Yongwen Xu. Literature review on web application gamication and analytics. Honolulu, HI, pages 11{05, 2011.

3. Conceptualization

3.1. Core Concepts:

- Context (Game Context, Non-Game Context)
- Game Design Element (Component, Dynamic, Mechanic)
- Gamification
- Motivation And Need Theory
- Player
- Player Model
- Player Type

3.2. Glossary of Terms

Table: Glossary of Terms

Term	Description
Ability	Possession of the means or skill to do something.

Achievements	Defined objectives
Alderfer's ERG Theory	E.R.G. theory assumes that a human being has three core needs that he strives to meet. They include obtaining his material existence needs, maintaining his interpersonal relatedness with significant other people, and seeking opportunities for his unique personal development and growth
Autonomy	Autonomy is the need to have independence and to be able make own choices. (Deci et al. 2010)
Avatar	A virtual representation of a player's character in a game. Common in role-playing games in which the player might take on the role of a magical creature or a medieval warrior.
A. Francisco-Aparicio's framework	A generic gamification framework proposed by A. Francisco-Aparicio (2013).
Badges	A visual token of a achievement. Usually designed to look like the real-world analogs, such as Boy Scout badges or the Good Housekeeping seal
Behavior-Change gamification	Seeks to form beneficial new habits among a population.
Belonging (social)	Need characterized by belonging to and being accepted by various groups. The social need represents striving for meaningful relationships. (Maslow 2003)
Boss fight	A difficult fight against a high-level opponent, called a boss. Often marks the end of a level or a section of a game
BrainHex	Typology of playing preferences motivated by combining existing findings from player research with neurobiological insights into the presumed underlying mechanisms

BrainHex Achiever	This archetype is more explicitly goal-oriented, motivated by long-term achievements.
BrainHex Conqueror	Players fitting the this archetype enjoy defeating impossibly difficult foes, struggling until they achieve victory, and beating other players.
BrainHex Daredevil	This play style is all about the thrill of the chase, the excitement of risk taking and generally playing on the edge.
BrainHex Mastermind	Players who fit this archetype enjoy solving puzzles and devising strategies, as well as focusing on making the most efficient decisions.
BrainHex Seeker	Motivated by interest mechanism, which relates to the brain area processing sensory information and memory association
BrainHex Socialiser	People are a primary source of enjoyment for players fitting this archetype
BrainHex Survivor	This archetype could be related to Lazzaro's serious fun, but in either case it can be seen as a special case of Caillois' ilinx, which purposefully courts controlled experiences of panic
B. Merino de Paz's framework	A generic gamification framework proposed by B. Merino de Paz (2013).
C. Li's framework	A business-specific gamification framework proposed by C. Li (2014)
C. Marache-Francisco and E. Brangier's framework	A generic gamification framework proposed by C. Marache-Francisco and E. Brangier (2013).
Challenges	Puzzles or other tasks that require effort to solve
Chance	Elements of randomness
Collections	Set of items or badges to accumulate

Combat	A defined battle, typically short-lived
Competence	Competence is the need to control the outcomes and experience a sense of ability (mastery). (White 1959)
Competition	One player or group wins, and the other loses
Constraints	Limitations or forced trade-offs
Content unlocking	Aspects available only when players reach objectives
Context	Context of gamification refers to the core service or an activity being gamified
Cooperation	Players must work together to achieve a shared goal
D. Robinson and V. Bellotti's framework	A generic gamification framework proposed by D. Robinson and V. Bellotti (2013)
Dignan, A's framework	A business-specific gamification framework proposed by Dignan, A (2011).
Domínguez, A.'s framework	A gamification framework in the learning context proposed by Domínguez, A. et al. (2013)
Duggan, K.'s framework	A business-specific gamification framework proposed by Duggan, K. et al. (2013)
Easy fun	Casual enjoyment, a way of blowing off steam without overly taxing yourself
Emotions	Curiosity, competitiveness, frustration, happiness, and so on
Engagement loop	the basic cycle of activity in a game, from motivation to activity to feedback,which in turn motivates further actions.
External gamification	Use of gamification to improve the relationships between businesses and customers, producing increased engagement, identification with the product,

	stronger loyalty, and ultimately higher revenues.
Extrinsic motivation	● Doing something for a reason other than for its own sake. ● The performance of an activity in order to attain some separable outcome. (Deci 2000)
Facilitator	Facilitator is a type of trigger. The goal of a facilitator is to trigger the behavior while also making the behavior easier to do. (Fogg 2009)
Feedback	Information about how the player is doing.
Flow Theory	States that in order for a task to be fully engaging it must reach an optimal mind state named flow, which is a state of optimal intrinsic motivation, full concentration, absorption and intense immersion. (Wu 2009; Xu 2011)
Fogg's Behavior Model	A behavior model for persuasive design (Fogg 2009)
Framework	Real or conceptual structure intended to serve as a support or guide for the building of something that expands the structure into something useful.
Game	● Free activity standing quite consciously outside of "ordinary" life, being "not serious", but at the same time intensely absorbing to the player. ● Non-serious but intensely engaging voluntary activities structured by rules and secretive social boundaries. (Huizinga 2000) ● A voluntary activity that operates within a "magic circle", in which

	players follow the rules of the game rather than those of the real world (Werbach, FTW)
GAME	A generic gamification framework proposed by A. Marczewski (2012).
Game atoms	Smallest parts of a game that can be isolated and studied individually.
Game component	A particular structure in a game, implementing the game's mechanics and dynamics. Points and badges are examples of game components
Game design	● Action of making sense of things related to a game (Mora. A et al. 2015) ● Act of deciding what a game should be (J. Schell 2008). ● The overall process of creating engaging games, based on an understanding of player desires, technological feasibility, and business objectives. Distinguished from the narrower term "game development" which is the technical implementation of a game (Werbach, FTW)
Game design fundamentals principles	1. Understanding design, systems, and interactivity, as well as player choice, action, and outcome. 2. Including a study of rule-making and rule-breaking, complexity and emergence, game experience, game representation, and social game interaction. 3. Adding the powerful connection between the rules of a game and the play that the rules engender, the pleasures games invoke, the meanings they construct, the

	ideologies they embody, and the stories they tell.
Game design methods	Game design-specific practices and processes
Game design patterns and mechanics	Commonly recurring parts of the design of a game that concern gameplay. (Deterding et al.(2011))
Game dynamics	● describe the effects of the mechanics on the subjective user experience over time and correspond to specific user motives. (Blohm & Leimeister. 2013) ● The conceptual structures underlying a game, such as the narrative and rules (constraints) that shape the game. These are the most abstract game elements. Players feel their effects but do not engage with them directly. (Werbach, FTW)
Game element	A design pattern that can be incorporated into a game. Game elements are the pieces that a game designer assembles in creating an engaging experience
Game interface design patterns	Common, successful interaction design components and design solutions for a known problem in a context, including prototypical implementations

Game design principles and heuristics	Evaluative guidelines to approach a design problem or analyze a given design solution
Game mechanics	● The processes that drive forward the action in a game, such as feedback or turns. Game mechanics are the actions that implement higher-level game dynamics and manifest themselves in lower-level game components. (Werbach, FTW) ● Building blocks for gamifying a core offer such as scoring systems or badges. Blohm & Leimeister (2013)
Game models	Conceptual models of the components of games or game experience
Game thinking	The process of addressing problems like a game designer, by looking at how to motivate players and create engaging, fun experiences. Sometimes called “gameful thinking”, in contrast to unstructured “playful thinking”
Gamefulness	● Game interface design patterns, game design patterns and mechanics, game design principles and heuristics, game models and game design methods. ● A systematic complement to “playfulness”. “Playfulness” broadly denotes the experiential and behavioral qualities of playing (paidia), “gamefulness” denotes the qualities of gaming (ludus). (Deterding et al. 2011)
Gamification	● Application of game-like accelerated user interface design to make electronic transactions both

	enjoyable and fast (N. Pelling 2011) ● Use of game design elements in non-game contexts. (S. Deterding et al. 2011) ● A process of enhancing a service with affordances for gameful experiences in order to support user's overall value creation (Hamari et al. 2012)
Gamified service bundles	A bundling of game-design elements into comprehensive, IT-based and increasingly ubiquitous enhancing services
Gifting	Opportunities to share resources with others
H. Jacobs' framework	A business-specific gamification framework proposed by H. Jacobs (2013)
Hard fun	A challenge or puzzle, which is fun because of the pleasure of overcoming it.
Hedax user type	A model that describes users' types in the gamification context.
Interest curve	The pattern of gradually increasing difficulty in a game, structured to keep users interested at every stage. (werbach, ftw)
Internal Gamification	The use of gamification to improve productivity within a organization in order to foster innovation, enhance camaraderie, or otherwise derive positive business results through their own employees.
Intrinsic motivation	● Doing something for its own sake. People are intrinsically motivated if they engage in activity without any hope of an external reward. ● The performance of an activity for the inherent satisfaction of the activity itself. (Deci 2000)
J. Kumar's framework	A business-specific gamification framework proposed by J. Kumar (2013).
K. Julius' framework	A business-specific gamification framework proposed by K. Julius (2013)
Leaderboard	A ranked list of participants in a game, with

	the highest scores on top.
Levels	Defined steps in player progression
Loyalty program	A program to reward regular customers with benefits in proportion to their level of activity.
Magic circle	The virtual or physical space where the rules of the game hold sway over those of the real world.
M. Versteeg's framework	A framework for moral persuasive gamification design proposed by M. Versteeg (2013).
MDA (Mechanics, Dynamics and Aesthetics)	A generic gamification framework proposed by R. Hunicke et al (2004).
Maslow's hierarchy of needs theory	States that a person has a pyramid hierarchy of needs that a person must satisfy from bottom to top. ()
Moral design framework	Framework for evaluating the ethics of a persuasive interaction in a traditional persuasive context.
Motivation	A disposition to do something you want (Mizoguchi)
Motivational affordances	The properties of an object that determine whether and how it can support one's motivational needs (Zhang 2008).
N. Kumar's framework	A business-specific gamification framework proposed by N. Kumar (2013).
Narrative	A consistent, ongoing storyline
Nicholson, S' framework	A generic gamification framework proposed by Nicholson, S. (2012)
Octalysis	A complete gamification framework proposed by Y.-k. Chou (2013).
P. Herzig's framework	A business-specific gamification framework proposed by P. Herzig (2014).
Persuasive Technology	Technology that is designed to change attitudes or behaviors of the users through persuasion and social influence.

Physiological Needs	The basic human needs required to sustain life and include food, clothing, and shelter. (Maslow 2003)
Planned Behavior Theory	A theory designed to predict and explain human behavior in specific contexts. (Ajzen 1985)
Player-Centered Design	A game design which is centered on an adaptive environment to feed the individual in players.
Points	Numerical representations of game progression
Progression	The player's growth and development
Progression stairs	The cycle of advancement through the levels or other steps in a game. More detailed version of the game's interest curve.
Pyramid of Gamification Elements	Developed by Werbach (2012), it is the basis for several other gamification design frameworks: mechanics, dynamics and components.
Quest	A specific mission or challenge for players of a game. The quest will usually have a narrative and an objective and a reward for completion.
Relatedness	Relatedness is the need to be connected with others, iterate with them, and experience caring for them. (Roy et al. 1995)
Relationships	Social interactions generating feelings of camaraderie, status, altruism, and so on
Resource acquisition	Obtaining useful or collectible items
Rewards	Benefits for some action or achievement
RAMP (Relatedness, Autonomy, Mastery, Purpose)	A motivation framework proposed by Marczewski (2012).
S. Reiss' framework	A business-specific gamification framework proposed by S. Reiss (2002).

Safe and Security	The need to be free of fear of physical danger, the need to be free of deprivation of basic physiological needs, and the need for self-preservation. (Maslow 2003)
Self-actualization	Represents the need to maximize one's potential and to become what one is capable of becoming (Hersey 1996)
Self-esteem	The need for esteem or recognition from others. (Maslow 2003)
Sgi	A generic gamification framework proposed by A. AlMarshedi et al (2015)
Signal	Signal is a type of trigger. The signal doesn't seek to motivate people or simplify the task. It just serves as a reminder. (Fogg 2009)
Simões, J's framework	A gamification framework in learning context proposed by Simões, J et al. (2013)
Six Steps to Gamification (6D)	A generic gamification framework proposed by K. Werbach et al (2012).
Skinner's reinforcement theory	Skinner's reinforcement theory or operant conditioning theory states that reinforced behaviors will tend to be repeated, while punished behavior tend to be decrease and will eventually end. (Geiser 2014)
Spark	Spark is a type of trigger. When a person lacks motivation to perform a target behavior, a trigger (spark) should be designed in tandem with a motivational element. (Fogg 2009)
Status	The rank or level of a player. Players are often motivated by trying to reach a higher level or status.
Success	A generic gamification framework proposed by D. DiTommaso. (2011).
Success model	An Information System success measurement synthesis in order to provide a guidance to the future.
Self Determination Theory	• An approach to human motivation

	and personality that uses traditional empirical methods while employing an organismic metatheory that highlights the importance of humans' evolved inner resources for personality development and behavioral self-regulation. (Ryan et al. 2000) ● A psychological theory developed by Edward Deci and Richard Ryan of the University of Rochester, along with many collaborators, which defines and emphasizes the importance of intrinsic motivation (Werbach, FTW)
Social graph	The network of relationships among friends, such as the matrix of connections on Facebook or other social networking sites.
Target Behavior	The expected behavior in the Fogg's Behavior Model
Teams	Defined groups of players working together for a common goal
Technology Acceptance Model (TAM)	A model which deals how users come to accept and use technology.
Transactions	Trading between players, directly or through intermediaries
Trigger	A trigger is something that tells people to perform a behavior now. (Fogg 2009)
Turns	Sequential participation by alternating players
Variable reward schedule	A prize or reward delivered on some non predictable basis. such as the payoff of a slot machine.
Virtual currency	A medium of exchange in a game, allowing players to purchase virtual goods or other benefits.
Virtual economy	A functional market system in a game, typically including virtual currency and virtual goods that are subject at least in part to economic forces.

Virtual goods	Virtual items that have value or uniqueness within a game environment. Players may be able to purchase virtual goods with virtual currency, real money, or through achievements within the game. Also called virtual assets.
Yee Motivational Component	
Yee Achievement	Motivational Component
Yee Advancement	Players who derive satisfaction from reaching goals, leveling quickly and accumulating in-game resources such as gold.
Yee Competition	Players who enjoy competing with other gamers on the battlefield or economy.
Yee Customization	Players who enjoy customizing the appearance of their characters.
Yee Discovery	Players who enjoy exploring the world and discovering locations, quests or artifacts that others may not know about.
Yee Escapism	Players who use the environment as a place to relax or relieve their stress from the real world.
Yee Immersion	Motivational Component
Yee Mechanics	Players who derive satisfaction from analyzing and understanding the underlying numerical mechanics of the system.
Yee Relationship	Players who are looking to form sustained, meaningful relationships with others.
Yee Role Playing	Players who enjoy being immersed in a story through the eyes of a character that they designed.
Yee Social	Motivational Component
Yee Socializing	Players who enjoy meeting and getting to know other gamers
Yee Teamwork	Players who enjoy working and collaborating with others.
Win state	the outcomes of a game that constitute

	“winning”. Typically defined by the rules of the game and the game’s feedback or rewards mechanisms.

3.2. Tree of Concepts

- Game
    - Game Design Element
        - Dynamics
            - Constraints
            - Emotions
            - Narrative
            - Progression
            - Relationships
        - Mechanics
            - Challenges
            - Chance
            - Competition
            - Cooperation
            - Feedback
            - Resource Acquisition
            - Rewards
            - Transactions
            - Turns
            - Win states
        - Components
            - Achievements
            - Avatars
            - Badges
            - Boss fights
            - Collections
            - Combat
            - Content unlocking
            - Gifting
            - Leaderboards
            - Levels
            - Points
            - Quests
            - Social Graph
            - Teams
            - Virtual Goals
            - Virtual Goods
- Context
    - Education
    - Health-care

        - Business
        - Social Networks
        - Sustainability
        - Orientation
        - Marketing
        - Crowdsourcing
        - Computer Science and Engineering
        - Research
        - Computer-supported cooperative work
- Theory
    - Motivation and Human Behavior
        - Fogg's Behavior Model
            - Targeted behavior
            - pro
            - Ability
            - Trigger
                - Facilitators
                - Spark
                - Signal
        - Need Theories
            - Maslow's Hierarchy of Needs
                - Physiological
                - Safety
                - Love/belonging
                - Esteem
                - Self-actualization
            - Alderfer's ERG
                - Existence
                - Relatedness
                - Growth
            - Self-determination
                - Autonomy
                - Relatedness
                - Competence
        - Skinner's Reinforcement (Instance Attributes: Positive, Negative_Escape, Negative_Active_Avoidance)
            - Reinforcement
            - Punishment
        - Csikszentmihalyi's Flow (Instance Attributes: Challenge, Ability)
    - Game Design Theories
        - Bartle's Player Types
            - Achiever
            - Socializer
            - Explorer

            - Killer
        - Yee’s Motivational Components
            - Achievement
                - Advancement
                - Mechanics
                - Competition
            - Social
                - Socializing
                - Relationship
                - Teamwork
            - Immersion
                - Discovery
                - Role-Playing
                - Customization
                - Escapism
        - Marczewski's Six-player Types
            - Achiever
            - Socializer
            - Philanthropist
            - Free-spirit
            - Player
                - Self-seeker
                - Networker
                - Consumer
                - Exploiter
            - Disruptor
                - Griefer
                - Destroyer
                - Influencer
                - Improver
- Player

3.4. Binary-Relations

Table y. Binary-relations table

Relation name	**consistsOf**
Source concept	**Dynamics**
Source cardinality	**(1,n)**
Target concept	**GDE_Mechanics**
Target cardinality	**(1,n)**

Inverse relation	**composes**
References	
Relation name	**representes**
Source concept	**Components**
Source cardinality	**(1,n)**
Target concept	**GDE_Mechanics**
Target cardinality	**(1,n)**
Inverse relation	**isReprentedBy**
References	
Relation name	**classifies**
Source concept	**Player_Model**
Source cardinality	**(1,n)**
Target concept	**Player**
Target cardinality	**(1,n)**
Inverse relation	**isClassifiedBy**
References	
Relation name	**isAppliedTo**
Source concept	**Gamification**
Source cardinality	**(1,n)**
Target concept	**Context**
Target cardinality	**(1,n)**
Inverse relation	**hasGamification**

References	
Relation name	**makeUseOf**
Source concept	**Gamification**
Source cardinality	**(1,n)**
Target concept	**Game_Design_Elements, Motivation_Behavior_Theories**
Target cardinality	**(1,n), (1,n)**
Inverse relation	**isUsedBy**
References	
Relation name	**composes**
Source concept	**GDE_Mechanics**
Source cardinality	**(1,n)**
Target concept	**Dynamics**
Target cardinality	**(1,n)**
Inverse relation	**consistsOf**
References	
Relation name	**isRepresentedBy**
Source concept	**GDE_Mechanics**
Source cardinality	**(1,n)**
Target concept	**Components**
Target cardinality	**(1,n)**
inverse relation	**represents**
References	

Relation name	**isClassifiedBy**
Source concept	**Player**
Source cardinality	**(1,n)**
Target concept	**Player_Type**
Target cardinality	**(1,n)**
Inverse relation	**classifies**
References	
Relation name	**hasGamification**
Source concept	**Context**
Source cardinality	**(1,n)**
Target concept	**Gamification**
Target cardinality	**(1,n)**
Inverse relation	**isAppliedTo**
References	
Relation name	**isUsedBy**
Source concept	**Game_Design_Elements, Motivation_Behavior_Theories**
Source cardinality	**(1,n), (1,n)**
Target concept	**Gamification**
Target cardinality	**(1,n)**
Inverse relation	**makesUseOf**
References	

4. Integration

At the moment this ontology was developed no other ontologies were found in order to integrate to this one.

5. Implementation

An implementation of this ontology is available at: